

Voice recognition technology

Voice recognition technology, also known as voice recognition (English: Automatic Speech Recognition, ARS) whose purpose is to let the machine "understand" human spoken language, contains two meanings: one is word for word in the human voice Content conversion word input into a computer-readable, such as buttons, binary code or sequence of characters; second spoken language is contained in the request or inquiry to be understood responses correctly, not all words of the proper conversion particular .

Speech recognition technology has developed to become involved in acoustics, linguistics, digital signal processing, statistical models, probability theory and information theory, vocal mechanism and hearing mechanism and artificial intelligence technologies a multidisciplinary integrated technology. Based speech recognition technology developed by modern speech recognition systems in many scenarios have been successful applications under different tasks the technology used will be different. Voice recognition system on the whole building process consists of two parts: training and recognition. Training is usually done offline, collect the mass of the pre-speech, language knowledge database for signal processing and mining, speech recognition system to obtain the desired "acoustic models" and "language model"; the recognition process is usually done online, the user real-time voice for automatic recognition. Identification process usually can be divided into "front-end" and "back-end" two modules: "front end" module is the role of endpoint detection (to remove excess silence and non-speech sound), noise reduction, feature extraction, etc.; "post end "of the module is the role of the trained" acoustic models "and" language model "eigenvector for the user to speak statistical pattern recognition (called" decoding "), which contains the text information obtained, in addition, the back-end module also there is an "adaptive" feedback module that can be customized for the user's voice learning, and thus the "acoustic model" and "voice model" necessary "correction" to further improve the recognition accuracy.

With the popularity of smart phones, speech technology more widely, including voice dialing, voice navigation; rapid popular micro-channel voice technology is one of the applications, micro-channel voice just simply save, transfer and playback, the machine does not involve the identification , I believe that in the near future will be universal and popular man-machine dialogue.

Mobile operators want to take advantage of a robot as a voice call, deal with simple issues such as balance inquiry, query package margin, check the latest deals and so on. Users only need to record a voice sent to this operator's customer service robot (through micro-channel public account other forms), the contents of the robot completed by voice response. This process is simplified as follows: in a recorded voice recognition unit need not be considered as a punctuate, without regard to the form returned to the user, only requires to be able to recognize the speech.

Please complete the following questions:

- 1, An established model illustrates all aspects of speech recognition technology
- 2, according to your model for the mobile company to develop a workable user operating rules
- 3, in order to validate your model use "balance bill" to check your model.